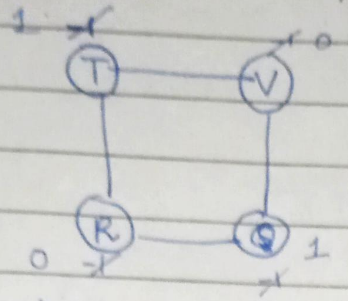
Date: Dec 2nd, 25 Lecture # 27 Day: Tuesd

Bipartite Graph



set#1 : {T, Q}

set#2 : {R, V}

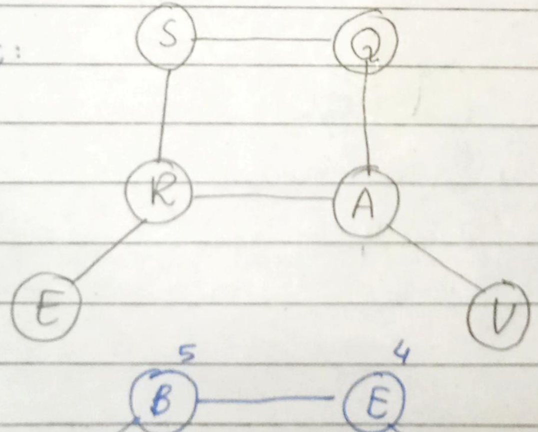
DFS traversal

for each v ∈ G.Adj[u]
    if (v.color == -1)
        v.color = 1 - u.color

if (visited & u.color == v.color)
    return false; // not a ~~Bipartite~~

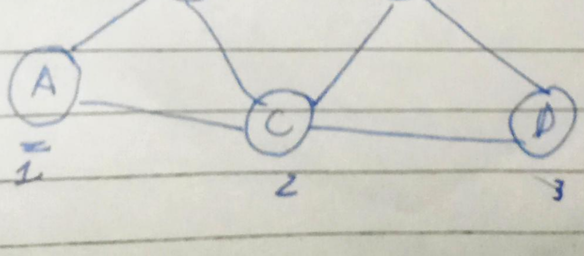
Hamiltonian Graph / cycle.

not :



'after visiting all vertices, if returns to the source.'

* Apply all possible paths using DFS



$\underbrace{\text{ACDEB}}_{\text{path}}$

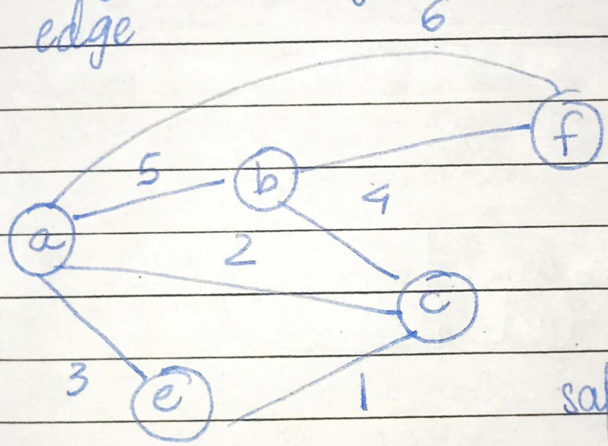
Applied DFS:

Date: Day:

```
if (vis_count == |G.V|)
    for each u ε G.Adj[V]
        if (u == src)
            // cycle exists
```

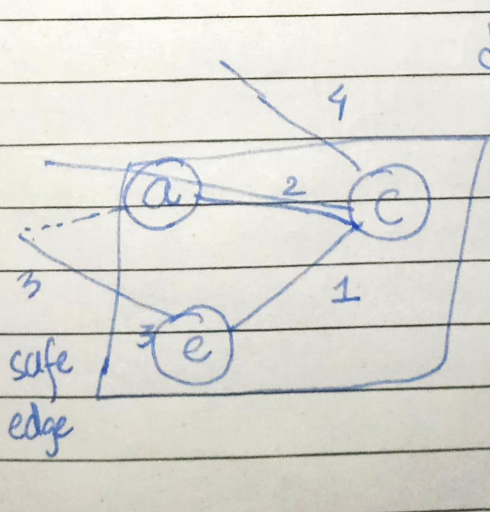
Lecture # 28 (Last)

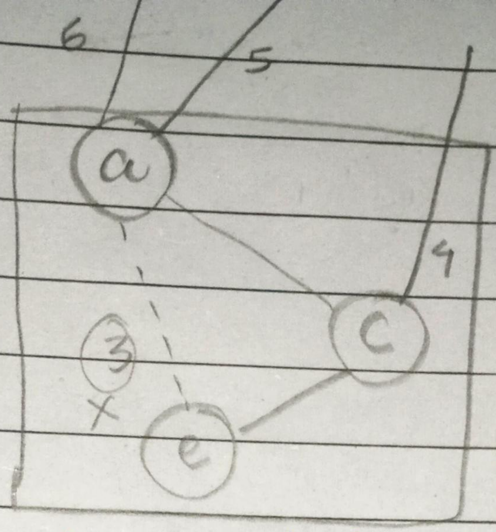
Light edge: outgoing edge, having min. cost.
Safe edge



safe edge:
The edge that doesn't make any cycle.

MST:





min
$\boxed{3}$ 4 5 6
light edge
but not safe

To create a MST : pick the Lighest safe edge.

safe edges
4, 5, 6

Q 8-

max jump

3	5	1	0	2
1	2	3	4	5

Find minimum hops req. to reach the last index.

```
K = 1
for ( i = A[k] - 1 ; i >= 0 ; i-- )
{
    j_size = A[k] - i
}
```

Q. Count all possible subset having sum equal to target value.

Tr value = 8

7	20	1	5	3	2	-12

{7, 1},
{-12, 8}
{5, 3}
{1, 5, 2}

```
if (trVal == 0)
    return 1;
if (N == 0)
    return 0;
```

$$\text{return } \underbrace{SS(N-1, trVal)}_{\text{exclude}} + \underbrace{SS(N-1, trVal - Arr[N])}_{\text{include}}$$

Good Luck Fellas!